

Qiyuan Wu

 qiyuanbillwu |  billwu67@ucla.edu |  424-407-6762

EDUCATION

- | | | |
|----------------|--|-----------------|
| 2024 - Present | University of California, Los Angeles
M.S. in Aerospace Engineering - Departmental Scholar | (GPA: 3.96/4.0) |
| 2021 - Present | University of California, Los Angeles
B.S. in Aerospace Engineering, Double Major in Applied Mathematics | (GPA: 3.96/4.0) |

RESEARCH EXPERIENCE

Viscous Streaming

Dec 2024 - Present

The SOFIA Laboratory, UCLA - Advisor: Prof. Jeff Eldredge

- Developing a computational framework for mean particle trajectories in viscous streaming flows over various geometries.
- Simplified the Navier–Stokes equations via small-amplitude expansion.
- Applied generalized Lagrangian mean theory to derive mean transport equations.
- Optimized computational efficiency by solving in the frequency domain.
- Solved partial differential equations using the Immersed Layers Method.

Lagrangian Coherent Structures (LCS)

June 2024 - Dec 2024

The SOFIA Laboratory, UCLA - Advisor: Prof. Jeff Eldredge

- Developed a computational framework for FTLE and LAVD fields to identify unsteady transport barriers in complex flows.
- Characterized unsteady saddle points using the FTLE field.
- Extracted the leading-edge vortex of a pitching flat plate using the LAVD field.
- Developed an open-source Julia package for FTLE/LAVD field computations ([ILMPostProcessing.jl](#)).

Autonomous UAV Motion Planning

June 2025 - Present

VECTR Laboratory, UCLA - Advisor: Prof. Brett Lopez

- Investigating aerodynamic-aware model predictive control of quadrotors.
- Bridging the gap between unsteady fluid dynamics, receding horizon control, and LCS for energy-efficient UAV motion planning.
- Optimized waypoint-based minimum-snap trajectories using gradient descent.
- Designed and implemented a framework for quadrotor planning, dynamics, control, and simulation.

Acoustically Coupled Combustion Instabilities

June 2024 - Sep 2024

Energy and Propulsion Research Laboratory, UCLA - Advisor: Prof. Ann Karagozian

- Investigated flame dynamics of single and coaxial methane diffusion jets at the pressure antinode of a standing acoustic wave.
- Captured high-speed flame responses across varied jet geometries, Reynolds numbers, frequencies, and pressure amplitudes.
- Classified flame dynamics into sustained oscillatory combustion (SOC), periodic lift-off and reattachment (PLOR), permanent lift-off (PLO), and blowoff (BO) using POD analysis.

PRESENTATIONS

Q. Wu. “*Numerical Solution of Flow Past an Oscillating Cylinder Using the Immersed Layers Method.*” Undergraduate Research Week, UCLA, May 2025. (Poster Presentation)

A. Hayrapetyan, D. Oviedo, **Q. Wu**, and A. Karagozian. “*Acoustically Coupled Single and Coaxial Fuel Jet Combustion at a Pressure Antinode.*” APS Division of Fluid Dynamics Meeting Abstracts, P34.004, November 2024. (Co-author; presented by A. Hayrapetyan)

Q. Wu. “*MLP-based Aerodynamic Coefficient Estimation of Flat Plate Airfoil in Transverse Gust*” MECH&AE 252E. Data Science for Fluid Dynamics, UCLA, June 2024. (Course Project Presentation)

GRADUATE COURSEWORK PROJECTS

Incompressible CFD with the Immersed-Boundary Projection Method

- Simulated lid-driven cavity flows ($Re = 400, 1000$); validated solver convergence on a staggered grid.
- Modeled flow past a cylinder with the discrete forcing approach.

Numerical Simulation of Supersonic Flow past a Cylinder

- Implemented Lax-Friedrichs and Steger–Warming schemes for Mach 4 cylinder flow.
- Validated pressure and Mach contours and bow shock location across grid resolutions.

AWARDS AND HONORS

UCLA Energy and Propulsion Research Laboratory Summer Scholarship June 2024

Department of Mechanical and Aerospace Engineering, UCLA

Departmental Scholar Program 2024 - Present

Department of Mechanical and Aerospace Engineering, UCLA

Dean’s Honors List 2021 - Present

EXTRACURRICULAR ACTIVITIES

Technical Vice President (2023 - 2024), CruX at UCLA 2021 - 2024

- Designed workshop series on building brain-computer interfaces (BCI).
- Led workshops for over 90 students on signal processing and machine learning in Python.
- Contributed to a BCI speller enabling brainwave-based typing; 1st Place, 2023 California Neurotechnology Conference.

Marine Mammal Rehabilitation Volunteer, California Wildlife Center 2022 - Present

- Rescued and rehabilitated over 100 sea lions, elephant seals, and other marine animals.